

Writing the First Image Loading & Displaying Program

Python Example (Using OpenCV and Matplotlib)

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python
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# Step 1: Import required libraries
import cv2
from matplotlib import pyplot as plt

# Step 2: Read an image (replace with the path of your image)
image = cv2.imread('sample_image.jpg')

# Step 3: Display image using OpenCV window
cv2.imshow('Displayed Image', image)
cv2.waitKey(0) # Wait until a key is pressed
cv2.destroyAllWindows()

# Step 4: Display image using Matplotlib (correct colors)
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
plt.imshow(image_rgb)
plt.title('Image Display using Matplotlib')
plt.axis('off')
plt.show()
```

Explanation :

- cv2.imread() → Reads the image.
- cv2.imshow() → Opens a window to display the image.
- cv2.waitKey(0) → Keeps the window open until a key is pressed.
- plt.imshow() → Displays image in notebook/plots with correct colors.